Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Schrödinger equation is equivalent to information conservation

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy.lean`
- Library: `UnitaryEntropy`
- Imports: `Mathlib.Analysis.SpecialFunctions.BinaryEntropy`, the Wigner library files `EqualMixtures.lean`, `Kadison.lean` and `Differentiable.lean` (which import the rest of `ForPhyslib/QuantumMechanics/Wigner`), and two new `ForMathlib` files:
  `ForMathlib/Analysis/Calculus/Deriv/Star.lean` (in a normed star algebra, the derivative of a path of unitaries through 1 is skew-adjoint, and $U^\dagger U$ is constant when $U' = XU$ with X skew-adjoint),
  `ForMathlib/Analysis/InnerProductSpace/Adjoint.lean` (in finite dimension, $U^\dagger U = 1$ already makes U unitary)
- Wigner library, in `ForPhyslib/QuantumMechanics/Wigner/`: `Density` (density operators, pure states `proj`, equal mixtures `mix`; moved there from this brief), `Basic`, `Normalize`, `Rigidity`, `Theorem` (Wigner's theorem without bijectivity, after Gehér, now in any

dimension, see Section 1.2; uniqueness up to one phase in finite dimension), `Projection` (the same for maps on rank-one projections), `Continuity` (Bargmann's triple product excludes the anti-unitary case near t_0), `Differentiable` (explicit, differentiable choice of the phase). Its `ForMathlib` helpers are `Analysis/Complex/ThreeCircles.lean`, `Analysis/InnerProductSpace/PhaseUnitary.lean` and `Analysis/InnerProductSpace/LinearMap.lean` (conjugate-linear isometries conjugate the inner product; polarization of $|x\rangle\langle y|$). Coordinate conjugation in an orthonormal basis came from TauCeti (`TauCeti.Analysis.InnerProductSpace.Conjugation`) until the library moved to Hilbert bases.

Setting. E is a finite-dimensional complex Hilbert space, and operators are continuous linear maps $E \to E$. A density operator (`IsDensity`) is a positive operator of trace 1. The pure state of a unit vector ψ is `proj` $\psi = |\psi\rangle\langle\psi|$ (Mathlib's `rankOne`), and `mix` $\rho\,\sigma = \frac{1}{2}\rho + \frac{1}{2}\sigma$. The von Neumann entropy `vonNeumann` is $\sum -\mu \log \mu$ over the roots μ of the characteristic polynomial. For self-adjoint ρ these are its eigenvalues (`vonNeumann_eq_sum`), and the definition makes invariance under conjugation immediate. `born` $\psi\,\phi = |\langle\psi|\phi\rangle|^2/(\|\psi\|^2\|\phi\|^2)$, and `mixEntropy` $p = H_2((1+\sqrt{p})/2)$ with H_2 Mathlib's `binEntropy`. The Schrödinger equation for a propagator is $U'(t) = (i\hbar)^{-1}H(t)\,U(t)$ with $H(t)$ self-adjoint.

Results.

Brief	Lean
Corollary 1, entropy of $\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$	`vonNeumann_mix_proj`, `mixEntropy_eq`
Corollary 1, strictly decreasing	`mixEntropy_strictAntiOn` (on $[0,1]$)
Corollary 1, $p(\phi\|\psi) = \lambda$ iff $S = \ldots$	`born_eq_iff` (for $\lambda \in [0,1]$)
Theorem 2 ($\Rightarrow$), unitary propagator	`star_mul_self_of_schrodinger`, `ContinuousLinearMap.mem_unitary_of_star_mul_self`
Theorem 2 ($\Rightarrow$), existence for constant H	`hasDerivAt_exp_schrodinger`: $U(t) = \exp(t(i\hbar)^{-1}H)$
Theorem 2 ($\Rightarrow$), existence for continuous $H(t)$	`exists_propagator_of_continuous`; strongly continuous $H(t)$: `exists_propagator_of_continuous_apply` (Section 1.3)
Theorem 2 ($\Rightarrow$), DR-INFO	`vonNeumann_unitary_conj`, `IsDensity.unitary_conj`, `schrodinger_conserves_entropy`
Theorem 2 ($\Leftarrow$), Step 1	`vonNeumann_eq_zero_iff`: $S(\rho) = 0$ iff ρ is pure
Theorem 2 ($\Leftarrow$), Steps 1–2 up to Wigner	`pure_and_born_of_preserves`, `proj_and_trace_of_preserves`
Theorem 2 ($\Leftarrow$), Step 2, Wigner's theorem	`Wigner.wigner_rankOne` (from `Wigner.wigner`)
Theorem 2 ($\Leftarrow$), anti-unitary case excluded	`Wigner.eventually_exists_unitary`, `PhaseEqOnSphere.false_of_unitary_of_antiunitary`
Theorem 2 ($\Leftarrow$), differentiable $U(t)$	`Wigner.exists_unitary_differentiableAt`
Theorem 2 ($\Leftarrow$), Step 3	`exists_hamiltonian` (from `star_eq_neg_of_hasDerivAt_of_mem_unitary`), for $\hbar \neq 0$
Theorem 2 ($\Leftarrow$), mixed states	`Wigner.affineOnDensities_of_map_mix`, `Wigner.AffineOnDensities.eq_conjOp`, `Wigner.mul_mul_star_eq_conjOp`
Theorem 2 ($\Leftarrow$), complete	`schrodinger_of_preserves`

Differences from the brief.

- This part is finite-dimensional; the brief has a separable Hilbert space. The calculus results (`star_mul_self_of_schrodinger`, `exists_hamiltonian`) hold in any Hilbert space for bounded $H(t)$. Corollary 1 and Theorem 2 through Wigner's theorem are redone in any dimension in Section 1.2.
- Corollary 1 is proved without diagonalizing the mixture. $\rho = \frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$ satisfies $\rho^3 = \rho^2 - \frac{1-\lambda}{4}\rho$ (`mix_proj_cubic`), so every eigenvalue is 0 or $(1 \pm \sqrt{\lambda})/2$. On these three points $-\mu \log \mu = \alpha\mu + \beta\mu^2$ for suitable α, β, and $\operatorname{tr}\rho = 1$, $\operatorname{tr}\rho^2 = (1+\lambda)/2$ fix the sum. This also covers the degenerate cases $\lambda = 0$ (double eigenvalue $\frac{1}{2}$) and $\lambda = 1$ (ρ pure).
- B EVMIX is used only for equal mixtures, which is all Step 2 needs. For the ($\Rightarrow$) direction, $\rho \mapsto U\rho U^\dagger$ is linear, so it preserves all mixtures.
- DR-INFO is $S(\rho_I) = S(\rho_F) = S(\rho_P)$. The reconstructed past $U^\dagger \rho_F U$ is ρ_I itself, which is also proved.

- The ($\Rightarrow$) direction shows that every propagator solving the equation is unitary. Existence is shown for constant H by the exponential, and for a continuous family of bounded $H(t)$ by the global existence theorem for linear ODEs, which the brief assumes (Section 1.3).

- `schrodinger_of_preserves` assumes $\Phi_{t_0} = \mathrm{id}$ and differentiability of $t \mapsto \Phi_t(\rho)$ at t_0 only on density operators, one ρ at a time. It concludes that $\Phi_t(|\psi\rangle\langle\psi|) = |U(t)\psi\rangle\langle U(t)\psi|$, and $\Phi_t(\rho) = U(t)\rho U(t)^\dagger$ for every density operator ρ, for t near t_0, with $U(t)$ unitary for every t, $U(t_0) = 1$ and $U'(t_0) = (i\hbar)^{-1}HU(t_0)$ for a self-adjoint H. This is the Schrödinger equation at t_0, which is what EVDIFF at t_0 gives. For a two-parameter evolution, one propagator solves it at every time (Section 1.1).

- Wigner's theorem is used in its non-bijective form (Gehér), so the bijectivity of the correspondence of pure states via the reconstructed past (Step 1) is not needed.

- $\Phi_t(\rho) = U(t)\rho U(t)^\dagger$ for mixed ρ follows from the pure states once Φ_t is affine, by the spectral decomposition of ρ (`Wigner.AffineOnDensities.eq_conjOp`). Affinity is not an extra assumption: a map of the density operators that preserves equal mixtures preserves all convex combinations of two of them (`Wigner.affineOnDensities_of_map_mix`), since it is bounded and preserves midpoints.

Problems found in the brief. No errors. Implicit points:

- The "iff" in Corollary 1 needs $\lambda \in [0, 1]$, where the entropy is strictly decreasing in λ; for other λ, $\sqrt{\lambda}$ and the logarithms are not meaningful.

- Step 3 assumes a norm-differentiable unitary family, which forces a bounded generator. On an infinite-dimensional Hilbert space it therefore covers bounded Hamiltonians only, and even then EVDIFF does not give such a family (Section 1.5). The brief "Multiple equivalent characterizations of the Schrödinger equation" treats the general case with Stone's theorem.

- In Step 3, $\hbar \neq 0$ is needed to write $A = H/(i\hbar)$.

- Wigner's theorem gives each $U(t)$ only up to a phase, and Step 3 differentiates $U(t)$. A differentiable choice exists. Extend Φ_t to all $|x\rangle\langle y|$ by polarization, so that it sends $|x\rangle\langle y|$ to $|Ux\rangle\langle Uy|$, and fix an orthonormal basis (b_k) and an index i. Then

$$U(t) = \left(\mathrm{Re}\,\langle b_i|\Phi_t(|b_i\rangle\langle b_i|)\,b_i\rangle\right)^{-1/2} \sum_k \Phi_t(|b_k\rangle\langle b_i|)\ |b_i\rangle\langle b_k|$$

  is the unitary with $\langle b_i|U(t)b_i\rangle > 0$. It is a finite expression in values of Φ_t on pure states, so it is differentiable at t_0 whenever these are.

- "Joint continuity and $\Phi_{t_0} = I$ exclude the anti-unitary case" needs only continuity at t_0 of $t \mapsto \Phi_t(P)$ for three pure states P: the imaginary part of Bargmann's triple product $\mathrm{tr}(\Phi_t(P_1)\Phi_t(P_2)\Phi_t(P_3))$ is $\frac{1}{4}$ at t_0 and would be $-\frac{1}{4}$ for an anti-unitary. In dimension 1 there is nothing to exclude.

1.1 The equation at every time

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/TwoParameter.lean`

- Imports: `SchroedingerEvolution.TwoParameter`, from the brief "Multiple equivalent characterizations of the Schrödinger equation" (for the two-parameter `IsEvolution` and `DrSceq` with $H(t)$; it imports this brief's main file), and the new `ForPhyslib/QuantumMechanics/Wigner/Propagator.lean` (sections of determinant one, their uniqueness up to a root of unity, and gluing along an interval).
- New `ForMathlib` file `ForMathlib/Analysis/Calculus/ContDiff/FiniteDimension.lean`: the determinant is a smooth function of the operator (Mathlib has only its continuity). `ForMathlib/Analysis/Calculus/Deriv/Star.lean` gains `star_mul_star_eq_neg_of_hasDerivAt_of_mem_unitary`: for a path of unitaries, $U'(t)U(t)^\dagger$ is skew-adjoint at every t. `Wigner.exists_unitary_differentiableAt` now also gives differentiability at every t near t_0 at which the family is differentiable.

Setting. E is finite-dimensional. A two-parameter evolution $\Phi_{t_0\to t_1}$, $t_0 \le t_1$ (`IsEvolution`), maps density operators affinely to density operators, with $\Phi_{t\to t} = I$, $\Phi_{t_1\to t_2}\circ\Phi_{t_0\to t_1} = \Phi_{t_0\to t_2}$, and $(\rho, t_0, t_1) \mapsto \Phi_{t_0\to t_1}(\rho)$ jointly continuous. There is no EVTI. DR-INFO (`DrInfo`) is $S(\Phi_{t_0\to t_1}(\rho)) = S(\rho)$ for all $t_0 \le t_1$. EVDIFF (`EvDiff`) is: for every reference time r and density operator ρ, $t \mapsto \Phi_{r\to t}(\rho)$ is differentiable at every $t > r$. DR-SCEQ (`DrSceq`) is conjugation, $\Phi_{t_0\to t_1}(\rho) = U(t_1)U(t_0)^\dagger\,\rho\,(U(t_1)U(t_0)^\dagger)^\dagger$, by a U with $U(0) = 1$ and $U'(t) = (i\hbar)^{-1}H(t)U(t)$ at every t, for self-adjoint $H(t)$.

Results.

Brief	Lean
DR-INFO, EVDIFF for $\Phi_{t_0\to t_1}$	`DrInfo`, `EvDiff`
Each $\Phi_{r\to t}$ is unitary	`exists_unitary`
Local unitaries of determinant one	`Wigner.eventually_isDetOneSection`
Uniqueness up to a constant root of unity	`Wigner.IsDetOneSection.exists_eq_smul`, `Wigner.eq_inner_smul_of_proj_eq`
One differentiable $U(t)$ on an interval	`Wigner.exists_isDetOneSection` (gluing: `IsPreconnected.exists_forall_of_eventually`), `exists_section`
One differentiable $U(t)$ for all t	`exists_propagator`
Step 3 at every time	`star_mul_star_eq_neg_of_hasDerivAt_of_mem_unitary`, `exists_hamiltonian_of_star_eq_neg`
Theorem 2, at every time	`drSceq_iff_drInfo_and_evDiff`

Differences from the brief.

- Finite dimension only; see the next paragraph for why the argument does not extend.
- EVDIFF asks for differentiability of $t \mapsto \Phi_{r\to t}(\rho)$ only at $t > r$, two-sided. There is no differentiability in the initial time and no one-sided derivative at $t = r$; a reference time $r < \tau$ gives the two-sided derivative of U at τ. A jump $H(t) = H_\beta$ for $t < 0$, $H(t) = H_\alpha$ for $t \ge 0$, with $H_\alpha - H_\beta$ not a multiple of 1, violates EVDIFF at $t = 0$ (the one-sided derivatives $(i\hbar)^{-1}[H_{\alpha,\beta}, \rho(0)]$ differ for some ρ), as it must: the U of DR-SCEQ is differentiable at every t.
- DR-INFO has no reconstructed past; in finite dimension it is not needed.

- The proof of ($\Leftarrow$). By Wigner's theorem each $\Phi_{r\to t}$ is induced by a unitary or an anti-unitary. Near any t' the maps $\Phi_{\min(t,t')\to\max(t,t')}$ are close to the identity and so unitary (`Wigner.eventually_exists_unitary`), and by the composition law $\Phi_{r\to t}$ has the type of $\Phi_{r\to t'}$; the type is locally constant, and $\Phi_{r\to r} = I$. Near $\tau > r$, conjugating by a unitary that induces $\Phi_{r\to\tau}$ makes the family the identity at τ, and the explicit `Wigner.unitaryOf` is differentiable near τ; multiplying by a branch of $\det(\cdot)^{-1/n}$, $n = \dim E$, gives determinant one. Two such families that induce the same maps differ by $c(t) = \langle W(t)e|V(t)e\rangle$, which is continuous with $c(t)^n = 1$, hence constant on a connected set. So the local families glue along (r,∞): the set of times reached from a fixed time is open and closed in (r,∞). The families for $r = -1, -2, \ldots$, normalized by $V(0)^\dagger$, agree, and give U; then $H(t) = i\hbar U'(t)U(t)^\dagger$.
- The propagator $U_{t_0\to t_1} = U(t_1)U(t_0)^\dagger$ satisfies the composition law exactly; no phase cocycle appears.

Problems found in the brief.

- The proof of Theorem 2 differentiates at one time, with phases chosen near that time. Going from "at each time" to one differentiable propagator for all times needs the local phase choices to be glued, and locally they are fixed only up to a continuous phase function. Fixing the determinant reduces this freedom to an n-th root of unity, which connectedness removes. The determinant needs finite dimension, so the infinite-dimensional case stays open.

1.2 Hilbert spaces of any dimension

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/InfiniteDimensional.lean`
- Library: `UnitaryEntropy`, which now builds every module `UnitaryEntropy.*`
- Imports:
  `ForPhyslib.QuantumMechanics.VonNeumannEntropy`,
  `ForPhyslib.QuantumMechanics.EqualMixtures` and `ForPhyslib.QuantumMechanics.Kadison` (equal mixtures give affinity, and the mixed-state extension),
  `ForMathlib.Analysis.InnerProductSpace.StrongDerivative`,
  `TauCeti.Analysis.InnerProductSpace.FunctionalCalculusComparison` (eigenvector action of the functional calculus, for the comparison in finite dimension),
  and the finite-dimensional part `UnitaryEntropy` (for `born`, `mixEntropy`, `mix_proj_cubic`, `mix_proj_sq`, `star_mul_self_of_schrodinger` and `exists_hamiltonian`, none of which assumes finite dimension)
- New file `ForPhyslib/QuantumMechanics/DensityOperator.lean`: density operators on a Hilbert space of any dimension, on top of the trace-class operators of PhyslibAlpha (`PhyslibAlpha/ProbabilisticTheory/HilbertSpace/TraceClass/`, from the physlib dependency; it builds against our Mathlib pin). Pure states, convex combinations, conjugation by an isometry, idempotent density operators are pure, and agreement with `IsDensity` in finite dimension.
- New file `ForPhyslib/QuantumMechanics/VonNeumannEntropy.lean`: the von Neumann entropy in $[0,\infty]$, its invariance under conjugation by an isometry, and $S(\rho) = 0$ iff ρ is pure.

- New `ForMathlib` material:
  `ForMathlib/Analysis/InnerProductSpace/PositiveTrace.lean` (the trace of a positive operator in $[0,\infty]$, independent of the Hilbert basis),
  `ForMathlib/Analysis/InnerProductSpace/HilbertBasisMaps.lean` (for a Hilbert basis: a vector is determined by its coordinates, Parseval for their moduli, the diagonal unitary $b_i \mapsto \delta_i b_i$, coordinatewise conjugation, the isometry $b_i \mapsto v_i$ onto an orthonormal family, and $V(V^\dagger x) = x$ when $\|V^\dagger x\| = \|x\|$),
  `ForMathlib/Analysis/InnerProductSpace/StrongDerivative.lean`
  (by Banach–Steinhaus, the strong derivative of an operator family is a bounded operator; for unitaries with $U(t_0) = 1$ it is skew-adjoint).

- The Wigner library `ForPhyslib/QuantumMechanics/Wigner/` (`Basic`, `Normalize`, `Rigidity`, `Theorem`) now works over a Hilbert basis of any Hilbert space. `Theorem` proves Wigner's theorem in any dimension, with isometries, and derives the finite-dimensional theorem from it. `Continuity` excludes the anti-unitary branch in any dimension, without a trace, and derives the finite-dimensional exclusion from it. `Differentiable` adds the differentiable choice of the phase in any dimension. The other files built on the library are unchanged.

Setting. E is a complex Hilbert space of any dimension; separability is not used. `posTrace` $T = \sum_i \langle e_i | T e_i \rangle \in [0,\infty]$ for a Hilbert basis (e_i). For positive T it does not depend on the basis (`posTrace_eq_tsum`): write $\langle e_i | R R^\dagger e_i \rangle = \|R^\dagger e_i\|^2$, expand by Parseval in a second basis and exchange the two sums of nonnegative terms. A density operator (`IsDensityOperator`) is a positive operator that is trace class in the sense of PhyslibAlpha (`IsTraceClass`) with `trace` $\rho = 1$; equivalently, $0 \le \rho$ and `posTrace` $\rho = 1$ (`isDensityOperator_iff`). The entropy is `vonNeumannEntropy` $\rho =$ `posTrace` $\eta(\rho)$ with $\eta(x) = -x \log x$ and $\eta(\rho)$ given by the continuous functional calculus. The spectrum of a density operator lies in $[0,1]$, where $\eta \ge 0$, so $\eta(\rho)$ is positive. `proj`, `mix`, `born` and `mixEntropy` are as above.

Results.

Brief	Lean
QST:H, density operators	`IsDensityOperator`, `isDensityOperator_proj`, `IsDensityOperator.mix`
QST:H, von Neumann entropy	`vonNeumannEntropy`
Agreement in finite dimension	`isDensityOperator_iff_isDensity`, `vonNeumannEntropy_eq_ofReal_vonNeumann`
Corollary 1, entropy of $\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi$	`vonNeumannEntropy_mix_proj`
Corollary 1, $p(\phi\|\psi) = \lambda$ iff $S = \dots$	`born_eq_iff_vonNeumannEntropy` (for $\lambda \in [0,1]$)
Theorem 2 ($\Rightarrow$), unitary conjugation preserves the entropy	`IsDensityOperator.vonNeumannEntropy_conj` (for isometries)
Theorem 2 ($\Rightarrow$), DR-INFO	`IsDensityOperator.conj`, `schrodinger_conserves_vonNeumannEntropy`
Theorem 2 ($\Leftarrow$), Step 1	`IsDensityOperator.vonNeumannEntropy_eq_zero_iff`
Theorem 2 ($\Leftarrow$), Steps 1–2 up to Wigner	`pure_and_born_of_preserves_vonNeumannEntropy`, `exists_isWignerMap_of_preserves`
Wigner's theorem, any dimension	`Wigner.IsWignerMap.exists_linearIsometry_or_conjLinearIsometry`, `Wigner.wigner_isometry`
Wigner's theorem, onto the rays	`Wigner.IsWignerMap.exists_unitary_or_antiunitary_of_surjective`
Theorem 2 ($\Leftarrow$), Steps 1–2 with Wigner	`exists_isometry_of_preserves`; `exists_unitary_of_preserves` if every pure state is an image
Theorem 2 ($\Leftarrow$), surjectivity from the reconstructed past	`exists_proj_of_past`, `exists_unitary_of_preserves_of_past`
Theorem 2 ($\Leftarrow$), anti-unitary case excluded	`eventually_exists_unitary_of_preserves` (from `Wigner.eventually_exists_unitary_of_unitary_or_antiunitary`)
Theorem 2 ($\Leftarrow$), surjectivity is needed	`shift_preserves_not_unitary`
Theorem 2 ($\Leftarrow$), differentiable $U(t)\psi$	`Wigner.exists_unitary_differentiableAt_apply`
Theorem 2 ($\Leftarrow$), Step 3	`ContinuousLinearMap.exists_hasDerivAt_apply`, `star_eq_neg_of_hasDerivAt_apply_of_mem_unitary`, `exists_hamiltonian_of_star_eq_neg`
Theorem 2 ($\Leftarrow$), mixed states	`affineOnDensityOperators_of_map_mix`, `AffineOnDensityOperators.eq_conjOp`
Theorem 2 ($\Leftarrow$), complete	`schrodinger_of_preserves_vonNeumannEntropy`

Differences from the brief.

- Wigner's theorem in any dimension follows Gehér's proof with a Hilbert basis (b_k). The first normalization changes: the images $f(b_k)$ are orthonormal but need not span, so they define an isometry V with $Vb_k = f(b_k)$, not a unitary. By Parseval, $\|V^\dagger f(v)\|^2 = \sum_k |\langle f(b_k)|f(v)\rangle|^2 = \sum_k |\langle b_k|v\rangle|^2 = 1 = \|f(v)\|^2$, so $f(v)$ lies in the range of V and $V^\dagger \circ f$ is a Wigner map that fixes every b_k. The diagonal unitary and the conjugation are then built from the basis as in finite dimension. The result is a linear or conjugate-linear isometry; it is onto, hence unitary or anti-unitary, when every ray is the image of a ray, and always in finite dimension.
- The anti-unitary branch is excluded by continuity without a trace. For $P = |a\rangle\langle a|$, Q,

R rank-one projections onto a, b, c and a fixed vector x, $\langle x|PQRP\,x\rangle = |\langle a|x\rangle|^2\,\Delta(a,b,c)$ (`inner_rankOne_mul_rankOne_mul_rankOne_mul_rankOne_apply`). Take the test triple $b_i, (b_i - b_k)/\sqrt{2}, (b_i - ib_k)/\sqrt{2}$ with $\Delta = (1+i)/4$, $x = b_i$, and the images $\Phi_t(P)$. The imaginary part is $\frac{1}{4}$ at t_0 and continuous at t_0 in operator norm, but ≤ 0 for an anti-unitary, since its triple product is $\overline{\Delta}$ and the factor $|\langle a|x\rangle|^2$ is nonnegative. The finite-dimensional `Wigner.eventually_exists_unitary`, which used $\mathrm{tr}(PQR) = \Delta$, is now a corollary. Continuity is assumed in operator norm; continuity in trace norm implies it.

- EVDIFF is formalized as in finite dimension: $t \mapsto \Phi_t(\rho)$ is differentiable at t_0 in operator norm for every density operator ρ. The conclusion is the Schrödinger equation on vectors, as in DR-SCEQ: $\psi(t) = U(t)\psi$ satisfies $\psi'(t_0) = (i\hbar)^{-1}H\psi(t_0)$ for every ψ, with one bounded self-adjoint H. It is not the equation for $t \mapsto U(t)$ in operator norm, which the finite-dimensional `schrodinger_of_preserves` gives; no choice of phases gives it (Section 1.5). The basis sum that fixes the phase in finite dimension does not converge in operator norm; instead, for a fixed unit vector e, Φ_t extended by polarization gives $(|Ux\rangle\langle Ue|)\,e = \langle Ue|e\rangle\,Ux$, and dividing by $|\langle Ue|e\rangle|$ fixes the phase one vector at a time. So $t \mapsto U(t)x$ is differentiable at t_0 for each x. By Banach–Steinhaus, applied to the difference quotients, the derivatives $x \mapsto (U(\cdot)x)'(t_0)$ form a bounded operator A. Differentiating $\langle U(t)x|U(t)y\rangle = \langle x|y\rangle$ gives $A^\dagger = -A$, and $H = i\hbar A$.

- Mixed states follow as in finite dimension, with the spectral sum replaced by the canonical expansion $\rho = \sum_i \mu_i\,|e_i\rangle\langle e_i|$, which may be infinite. Write $\rho = m\rho_F + (1-m)\sigma_F$ with ρ_F a finite part of weight m; Φ_t and $U(t)\cdot U(t)^\dagger$ agree on ρ_F by affinity, so they differ on ρ by at most $2(1-m)$ in operator norm, and $m \to 1$ (`AffineOnDensityOperators.eq_conjOp`). Affinity again comes from equal mixtures (`affineOnDensityOperators_of_map_mix`).

- No spectral decomposition is used. The entropy is the trace of $\eta(\rho)$, and pure states are found through the spectrum: $S(\rho) = 0$ forces $\eta(\rho) = 0$, so η vanishes on the spectrum, which then lies in $\{0,1\}$; ρ is idempotent, and an idempotent density operator is $|\psi\rangle\langle\psi|$ (`IsDensityOperator.eq_proj_of_isIdempotentElem`).

- Corollary 1 uses the cubic relation as in finite dimension, with the spectral mapping theorem for polynomials in place of eigenvalues. Then $\eta(\rho) = \alpha\rho + \beta\rho^2$, and its trace is computed from $\sum_i \langle e_i|x\rangle\langle y|e_i\rangle = \langle y|x\rangle$ for the rank-one terms.

- The ($\Rightarrow$) direction uses only $U(t)^\dagger U(t) = 1$, which every propagator satisfies. Conjugation $X \mapsto UXU^\dagger$ by an isometry is a non-unital star algebra homomorphism, so it commutes with the functional calculus of η (since $\eta(0) = 0$), and it preserves `posTrace`. Unitarity is not needed.

- Entropies are compared in $[0, \infty]$; in infinite dimension $S(\rho) = \infty$ occurs.

- DR-INFO is formalized in full: $S(\Phi\rho) = S(\rho)$, and every density operator ρ has a reconstructed past, a density operator σ with $\Phi\sigma = \rho$; then $S(\sigma) = S(\rho)$ by the first condition. The finite-dimensional part does not need the past, and `exists_isometry_of_preserves` shows what remains without it.

Problems found in the brief.

- In infinite dimension the reconstructed past in DR-INFO is essential, and Step 1 uses it when it says that pure states "correspond bijectively". DR-INFO presupposes a past ensemble ρ_P for every initial ensemble, so Φ is onto the density operators, as in the introduction's

"deterministic and reversible". Then every pure state is the image of a pure state, since its past has entropy zero (`exists_proj_of_past`), and Wigner's theorem gives a unitary or an anti-unitary (`exists_unitary_of_preserves_of_past`). The brief's proof is therefore correct, but the past cannot be dropped. Conjugation $\rho \mapsto S\rho S^\dagger$ by the shift S on $\ell^2(\mathbb{N})$ maps density operators to density operators and preserves the entropy and equal mixtures, but no pure state goes to $|e_0\rangle\langle e_0|$. So it is induced by no unitary and by no anti-unitary (`shift_preserves_not_unitary`), and Wigner's theorem gives only an isometry. In finite dimension the past is automatic.

- The time structure of [B]EVMIX does not replace the past. On $L^2(0,\infty)$ the right shifts $(V_t f)(x) = f(x-t)$ for $x > t$ and 0 otherwise are isometries forming a strongly continuous semigroup with $V_0 = I$. Then $\Phi_{t_0\to t_1}(\rho) = V_{t_1-t_0}\rho V_{t_1-t_0}^\dagger$ satisfies [B]EVMIX, preserves the entropy, and is induced by no unitary for $t_1 > t_0$; it has no reconstructed past. (By the Wold–Cooper decomposition, every strongly continuous semigroup of isometries is a unitary group plus copies of this one.) The example is formalized as `shift_semigroup_counterexample` in the brief "Multiple equivalent characterizations of the Schrödinger equation"; the Wold–Cooper decomposition is not. A norm-continuous semigroup of isometries is $V_t = e^{tA}$ with bounded A, and differentiating $V_t^\dagger V_t = I$ at 0 gives $A^\dagger = -A$, so it is unitary. The example therefore needs an unbounded generator, which Step 3 does not cover anyway.

- EVDIFF for every density operator forces a bounded Hamiltonian, as the Banach–Steinhaus step shows. Unbounded Hamiltonians, whose evolution is differentiable only on a dense set of states, are outside the theorem as stated; see Section 1.6. Otherwise no problems beyond those of the finite-dimensional part.

1.3 The propagator of a time-dependent Hamiltonian

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/Propagator.lean`

- Imports: `UnitaryEntropy.InfiniteDimensional` (for `schrodinger_conserves_vonNeumannEntropy`), `UnitaryEntropy.TwoParameter` (for the two-parameter `IsEvolution`, `DrSceq`, `DrInfo`, `EvDiff` and `drSceq_iff_drInfo_and_evDiff`), and the new `ForMathlib.Analysis.ODE.Linear` and `ForMathlib.Analysis.InnerProductSpace.HilbertSchmidt`.

- New `ForMathlib` file `ForMathlib/Analysis/ODE/Linear.lean`: global existence and uniqueness for $x' = L(t)x$ with L a strongly continuous family of bounded operators on a Banach space (`exists_hasDerivAt_clm_apply`, `eq_of_hasDerivAt_clm_apply_of_continuous_apply`; `eq_of_hasDerivAt_clm_apply` for continuous L), the strong propagator of such a family, invertible at all times (`exists_hasDerivAt_apply`, `eq_of_hasDerivAt_apply`, `isUnit_of_hasDerivAt_apply`), the same for $U' = A(t)U$ in a Banach algebra (`exists_hasDerivAt_mul`, `eq_of_hasDerivAt_mul`), and unitarity of the solution for skew-adjoint $A(t)$ (`mul_star_self_eq_one_of_hasDerivAt`, `exists_hasDerivAt_mul_mem_unitary`). It imports Mathlib's `Mathlib.Analysis.ODE.ExistUnique` (Picard–Lindelöf, uniqueness, Grönwall), `Mathlib.Analysis.ODE.Gronwall`, `Mathlib.Analysis.Normed.Operator.BanachSteinhaus` and `ForMathlib/Analysis/Calculus/Deriv/Star.lean`.

- New `ForMathlib` file `ForMathlib/Analysis/InnerProductSpace/HilbertSchmidt.lean`: composing with a Hilbert–Schmidt operator S ($\operatorname{tr} SS^\dagger < \infty$) turns bounded strong convergence into norm convergence (`ContinuousLinearMap.tendsto_mul_of_tendsto_apply`, from $\|R\|^2 \le \sum_i \|Re_i\|^2$, `ContinuousLinearMap.sq_norm_le_tsum_norm_apply_sq`), so a strongly differentiable, norm-Lipschitz family $F(t)$ gives a norm-differentiable $F(t)S$ (`ContinuousLinearMap.hasDerivAt_mul_of_hasDerivAt_apply`). It imports `ForMathlib/Analysis/InnerProductSpace/PositiveTrace.lean`, `Mathlib.Analysis.Calculus.Deriv.Slope` and `Mathlib.Analysis.Normed.Group.Tannery`.

Setting. E is a complex Hilbert space of any dimension, and $H : \mathbb{R} \to \mathcal{B}(E)$ is continuous in operator norm with every $H(t)$ self-adjoint. The brief's Theorem 2 ($\Rightarrow$) starts from "The Schrödinger equation gives rise to a differentiable unitary propagator $U_{t_0 \to t_1}$". Here the propagator is constructed. The second half of the section assumes only strong continuity: $t \mapsto H(t)\psi$ is continuous for every ψ.

Results.

Brief	Lean
Linear ODE, global existence and uniqueness	`exists_hasDerivAt_clm_apply`, `eq_of_hasDerivAt_clm_apply`
Propagator in a Banach algebra; unitary for skew-adjoint $A(t)$	`exists_hasDerivAt_mul`, `exists_hasDerivAt_mul_mem_unitary`
Theorem 2 ($\Rightarrow$), the propagator exists and is unitary	`exists_propagator_of_continuous`
Theorem 2 ($\Rightarrow$), the propagator is unique	`propagator_unique`
Theorem 2 ($\Rightarrow$), DR-SCEQ $\Rightarrow$ DR-INFO with this propagator	`exists_propagator_conserves_vonNeumannEntropy`
Two-parameter evolution of a propagator	`TwoParameter.conjEvolution`, `TwoParameter.isEvolution_conjEvolution`
Theorem 2 ($\Rightarrow$), two-parameter, finite dimension: QST:H + [B]EVMIX, DR-SCEQ, DR-INFO, EVDIFF	`TwoParameter.exists_propagator_isEvolution`
Strong propagator of a strongly continuous $L(t)$: existence, uniqueness, invertibility	`exists_hasDerivAt_apply`, `eq_of_hasDerivAt_apply`, `isUnit_of_hasDerivAt_apply`
Theorem 2 ($\Rightarrow$), strongly continuous $H(t)$: the propagator exists, is unitary and unique	`exists_propagator_of_continuous_apply`, `propagator_unique_of_continuous_apply`
Theorem 2 ($\Rightarrow$), strongly continuous $H(t)$: DR-SCEQ $\Rightarrow$ DR-INFO	`exists_propagator_conserves_vonNeumannEntropy_of_continuous_apply`
Theorem 2 ($\Rightarrow$), strongly continuous $H(t)$: EVDIFF, $U(t)\rho U(t)^\dagger$ solves the von Neumann equation in operator norm	`hasDerivAt_conj_of_continuous_apply`, `exists_propagator_hasDerivAt_conj_of_continuous_apply`
EVDIFF for the two-parameter evolution, any dimension	`TwoParameter.differentiableAt_conjEvolution`

Proof.

- Mathlib's Picard–Lindelöf theorem solves $x' = f(t,x)$ on $[t_0 - T, t_0 + T]$ only when $\|f\| \le L$ on a ball of radius a with $LT \le a$. For the linear field $f(t,x) = L(t)x$, L grows with a, so this gives only $T < 1/\sup\|L(t)\|$. On $[t_0 - T, t_0 + T]$ the equation is therefore solved for the truncated field $L(t)\pi_R(x)$, where $\pi_R(x) = \min(1, R/\|x\|)\,x$ is the radial retraction onto the ball of radius R. It is bounded by MR, with $M = \sup_{[t_0-T,t_0+T]}\|L(t)\|$, and globally 2-Lipschitz, so Picard–Lindelöf applies for every T. Since $\|\pi_R(x)\| \le \|x\|$, Grönwall's inequality on both sides of t_0 (the left side by reflection $t \mapsto 2t_0 - t$) gives $\|x(t)\| \le \|x_0\|e^{MT} < R$ for $R = \|x_0\|e^{MT} + 1$. So $\pi_R(x(t)) = x(t)$ and x solves the linear equation.

- The interval solutions agree on common intervals by Mathlib's uniqueness theorem (`ODE_solution_unique_of_mem_Icc`, Lipschitz constant M), and $x(t) = x_{|t-t_0|+1}(t)$ is a global solution: near each t it agrees with one interval solution, which is differentiable at the interior point t.

- The propagator is the case $L(t)U = A(t)U$, $A(t) = (i\hbar)^{-1}H(t)$, in the Banach algebra $\mathcal{B}(E)$. $U^\dagger U$ is constant (`star_mul_self_eq_of_hasDerivAt`), hence 1. For $UU^\dagger$, which has no such conservation law, $D = UU^\dagger - 1$ solves the linear equation $D' = AD - DA$, since $A^\dagger = -A$, with $D(t_0) = 0$; by uniqueness $D = 0$. So $U(t)$ is unitary in any dimension, not only an isometry.

- For the two-parameter evolution $\Psi_{t_0\to t_1}(\rho) = W\rho W^\dagger$, $W = U(t_1)U(t_0)^\dagger$, the composition law is $U(t_2)U(t_1)^\dagger U(t_1)U(t_0)^\dagger = U(t_2)U(t_0)^\dagger$, and joint continuity follows from the continuity of U. DR-SCEQ then holds by definition, and DR-INFO and EVDIFF follow from `TwoParameter.drSceq_iff_drInfo_and_evDiff`.

- Strongly continuous $L(t)$. Operator-norm continuity was used only twice: for the continuity of $t \mapsto L(t)x$ at fixed x, which is all that Mathlib's Picard–Lindelöf theorem asks for, and for a bound on $\|L(t)\|$ over a compact interval. That bound now comes from the Banach–Steinhaus theorem (`exists_norm_le_of_continuous_apply`), since $t \mapsto \|L(t)x\|$ is bounded on the interval for each x. The rest of the vector-level proof is unchanged.

- The operator $U(t)$ is assembled from the vector solutions: $U(t)x = x(t)$ for the solution with $x(t_0) = x$. It is linear (over $\mathbb{C}$, for $\mathbb{C}$-linear $L(t)$) by uniqueness, and bounded by Grönwall, $\|U(t)x\| \le e^{M|t-t_0|}\|x\|$. Only $t \mapsto U(t)x$ is differentiable, for each x. $U(s)$ is invertible: if V is the propagator with $V(s) = 1$, uniqueness gives $V(t)U(s) = U(t)$ and $U(t)V(t_0) = V(t)$, so $V(t_0)$ is a two-sided inverse of $U(s)$.

- Unitarity for $L(t) = (i\hbar)^{-1}H(t)$: $\langle U(t)\psi, U(t)\phi\rangle$ has derivative $\langle U\psi, AU\phi\rangle + \langle AU\psi, U\phi\rangle = 0$ since $A^\dagger = -A$, so $U(t)^\dagger U(t) = 1$ (`star_mul_self_of_hasDerivAt_apply`). An isometry with a right inverse V is unitary: $V = U^\dagger UV = U^\dagger$.

- EVDIFF. U is not differentiable in operator norm, but it is Lipschitz: with $\|A(r)\| \le M$ on $[t-1, t+1]$ (Banach–Steinhaus) and $\|U(r)\psi\| = \|\psi\|$, the mean value theorem gives $\|U(s) - U(t)\| \le M|s-t|$ (`exists_eventually_norm_sub_le`). Write $\rho = SS^\dagger$ with $S = \sqrt{\rho}$, so $\operatorname{tr} SS^\dagger = 1$. The difference quotients $D_s = (U(s) - U(t))/(s-t) - A(t)U(t)$ are bounded and tend to 0 strongly, and $\|D_sS\|^2 \le \sum_i \|D_sSe_i\|^2 \to 0$ by dominated convergence, with dominating sequence $C^2\|Se_i\|^2$. So $G(s) = U(s)S$ has the norm derivative $A(t)U(t)S$, and the product rule for $\rho(s) = G(s)G(s)^\dagger$ gives $\rho' = A\rho + \rho A^\dagger = (i\hbar)^{-1}[H(t), \rho]$ (`hasDerivAt_conj_of_continuous_apply`). For the two-parameter evolution, $\Psi_{r\to t}(\rho) = U(t)\sigma U(t)^\dagger$ with the density operator $\sigma = U(r)^\dagger \rho U(r)$ (`TwoParameter.differentiableAt_conjEvolution`).

Differences from the brief.

- $H(t)$ is assumed bounded and continuous in operator norm. For unbounded $H(t)$ the propagator needs Kato-type hypotheses and is not constructed (open problems). A family of bounded $H(t)$ that is only strongly continuous has a strongly differentiable unitary propagator (`exists_propagator_of_continuous_apply`). In general it is not differentiable in operator norm, so the Schrödinger equation cannot be stated for U itself: on $L^2[0,1]$, let $H(t)$ multiply by the indicator of $[0,t]$. Then $\|(H(t)-H(s))\psi\|^2 = \int_s^t |\psi|^2 \to 0$ but $\|H(t) - H(s)\| = 1$ for $s \neq t$ in $[0,1]$, and $U(t)$ multiplies by $e^{-i(t-x)_+/\hbar}$, whose difference quotient differs from $(i\hbar)^{-1}H(t)U(t)$ by at least $1/(\pi\hbar)$ in norm, from the values at x just above t (`indicatorHamiltonian_counterexample`, Section 1.4). EVDIFF holds nevertheless, for the states: $t \mapsto U(t)\rho U(t)^\dagger$ is differentiable in operator norm for every density operator ρ (`hasDerivAt_conj_of_continuous_apply`). The two-parameter structure `IsEvolution` and `drSceq_iff_drInfo_and_evDiff` are finite-dimensional and are not applied in this case.

Problems found in the brief. None: for bounded continuous $H(t)$ the assumed propagator exists, is unique and is unitary.

1.4 A strongly continuous Hamiltonian on $L^2[0,1]$

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/IndicatorHamiltonian.lean`
- Imports: `UnitaryEntropy.Propagator` (for `propagator_unique_of_continuous_apply`) and the new `ForMathlib.MeasureTheory.Function.LpSpace.Multiplication`.
- New `ForMathlib` file `ForMathlib/MeasureTheory/Function/LpSpace/Multiplication.lean`: the multiplication operator `Lp.mulL` g on $L^p(\mu)$ for an essentially bounded g, built from Mathlib's `ContinuousLinearMap.holderL`, with its pointwise formula (`Lp.coeFn_mulL`, `Lp.eq_mulL`); $\|\mathrm{mulL}\, g\| \le C$ if $|g| \le C$ a.e. (`Lp.norm_mulL_le`) and $\ge c$ if $|g| \ge c$ on a set of positive finite measure (`Lp.le_norm_mulL`); strong convergence when the multipliers are uniformly bounded and converge a.e. (`Lp.tendsto_mulL_apply`, from dominated convergence for the L^p norm, `tendsto_eLpNorm_of_dominated`); self-adjointness on L^2 for real g (`Lp.isSelfAdjoint_mulL`). It imports `Mathlib.Analysis.InnerProductSpace.Adjoint`, `Mathlib.MeasureTheory.Function.Holder` and `Mathlib.MeasureTheory.Function.L2Space`.

Setting. $L^2[0,1]$ is L^2 of Lebesgue measure restricted to $[0,1]$, on $\mathbb{R}$. $H(t)$ (`indicatorHamiltonian`) multiplies by the indicator of $[0,t]$, and $U(t)$ (`indicatorPropagator`) by $e^{-i(t-x)_+/\hbar}$ (`indicatorPhase`). This is the example in Section 1.3, "Differences from the brief".

Results.

Statement	Lean
$H(t)$ is self-adjoint with $\|H(t)\| \le 1$	`isSelfAdjoint_indicatorHamiltonian`, `norm_indicatorHamiltonian_le`
$t \mapsto H(t)\psi$ is continuous for every ψ	`continuous_indicatorHamiltonian_apply`
$\|H(t) - H(s)\| = 1$ for $s \ne t$ in $[0,1]$; H is norm-continuous at no $t \in (0,1)$	`norm_indicatorHamiltonian_sub`, `not_continuousAt_indicatorHamiltonian`
$U(0) = 1$, and $\frac{d}{dt}U(t)\psi = (i\hbar)^{-1}H(t)U(t)\psi$ for every ψ	`indicatorPropagator_zero`, `hasDerivAt_indicatorPropagator_apply`
For $\hbar > 0$, U is norm-differentiable at no $t \in (0,1)$	`not_differentiableAt_indicatorPropagator`
H satisfies the hypotheses of `exists_propagator_of_continuous_apply`, and no propagator V with $V(t_0) = 1$ is norm-differentiable at any $t \in (0,1)$	`indicatorHamiltonian_counterexample`

Proof.

- Strong continuity: for $x \ne t$, $\mathbf{1}_{x \le s} \to \mathbf{1}_{x \le t}$ as $s \to t$, and $\{t\}$ is a null set. The multipliers are bounded by 1, so dominated convergence gives $H(s)\psi \to H(t)\psi$.
- For $s < t$, $H(t) - H(s)$ multiplies by the indicator of $(s,t]$, which is bounded by 1 and equals 1 on a set of measure $t - s > 0$.
- Strong derivative: for $x \ne t$, $s \mapsto e^{-i(s-x)_+/\hbar}$ has derivative $(i\hbar)^{-1}\mathbf{1}_{x \le t}\, e^{-i(t-x)_+/\hbar}$ at t. Its difference quotients are bounded by $1/|\hbar|$, since $u \mapsto u_+$ and $\theta \mapsto e^{i\theta}$ are 1-Lipschitz, and dominated convergence applies again. This holds for every real $\hbar$.
- No norm derivative: a norm derivative of U at t would also be the strong derivative $(i\hbar)^{-1}H(t)U(t)$, which multiplies by 0 on $(t,1]$. For $t < s < \min(t + \hbar, 1)$ and $x \in (t, (t+s)/2)$, the difference quotient multiplies by $(e^{-i(s-x)/\hbar} - 1)/(s - t)$. Its modulus is $2\sin((s-x)/2\hbar)/(s-t) \ge 1/(\pi\hbar)$, by $\sin z \ge 2z/\pi$ on $[0, \pi/2]$. So the norm of the difference is at least $1/(\pi\hbar)$ as $s \downarrow t$.
- Every propagator: if $V(t_0) = 1$, then $s \mapsto U(s)U(t_0)^{-1}$ solves the same equation, where $U(t_0)^{-1}$ multiplies by the inverse phase. So $V(s) = U(s)U(t_0)^{-1}$ by `propagator_unique_of_continuous_apply`. If V were norm-differentiable at t, so would be $U(s) = V(s)U(t_0)$.

Notes. EVDIFF holds for this H (`hasDerivAt_conj_of_continuous_apply`), so the states $U(t)\rho U(t)^\dagger$ are norm-differentiable while the propagator is not.

1.5 No norm-differentiable propagator

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/NormDifferentiable.lean`
- Imports: `ForPhyslib.QuantumMechanics.VonNeumannEntropy` and `ForPhyslib.QuantumMechanics.Kadison` (density operators, entropy, products of rank-one projections) and `ForPhyslib.QuantumMechanics.Wigner.Propagator` (two unitaries with the same action on pure states differ by a phase, `Wigner.eq_inner_smul_of_proj_eq`).

Setting. In any dimension, `schrodinger_of_preserves_vonNeumannEntropy` chooses the phases of the unitaries $U(t)$ so that $t \mapsto U(t)\psi$ is differentiable at t_0 for every ψ (Section 1.2). In finite dimension the same hypotheses make $t \mapsto U(t)$ differentiable in operator norm. In infinite dimension no choice of phases does. On $\ell^2(\mathbb{N})$ with basis (e_k) let

$$U(t) = 1 + (e^{it} - 1)\, |e_{k(t)}\rangle \langle e_{k(t)}|\,, \qquad k(t) = \lceil 1/|t| \rceil + 1,$$

which multiplies the component along $e_{k(t)}$ by e^{it} (`movingPhase`), and $\Phi_t(\rho) = U(t)\rho U(t)^\dagger$. At $t = 0$ the factor $e^{it} - 1$ vanishes, so $U(0) = 1$.

Results.

Statement	Lean
$1 + (z-1)\,\|e\rangle\langle e\|$ is unitary for $\|z\| = 1$	`phaseOp_mem_unitary`
$\|UρU^\dagger - ρ\| \le (2\|z-1\| + \|z-1\|^2)\,\|ρe\|$ for self-adjoint $ρ$	`norm_phaseOp_mul_mul_star_sub_le`
$\|ρe_n\| \to 0$ for a density operator $ρ$	`tendsto_norm_apply_hilbertBasis`
EVDIFF at 0 in operator norm, with derivative 0	`hasDerivAt_movingPhase_conj`
No unitary family inducing Φ_t near 0 is norm-differentiable at 0	`not_exists_differentiableAt_movingPhase`
All hypotheses of `schrodinger_of_preserves_vonNeumannEntropy` hold, and no norm-differentiable $U(t)$ exists	`movingPhase_counterexample`

Proof.

- Φ_t is conjugation by a unitary, so it maps density operators to density operators, preserves the entropy and equal mixtures, and $\sigma = U(t)^\dagger \rho\, U(t)$ is a reconstructed past.
- EVDIFF at 0: with $P = |e_k\rangle\langle e_k|$ and $a = e^{it} - 1$, $\Phi_t(\rho) - \rho = a\,P\rho + \bar{a}\,\rho P + |a|^2 P\rho P$, and $\|P\rho\| = \|\rho P\| = \|\rho e_k\|$. So $\|\Phi_t(\rho) - \rho\| \le (2|a| + |a|^2)\|\rho e_{k(t)}\|$, where $|a|/|t| \to 1$ and $\|\rho e_{k(t)}\| \to 0$ as $t \to 0$: by $\|\rho e_n\| \le \|\sqrt{\rho}\|\,\|\sqrt{\rho}\, e_n\|$ and $\sum_n \|\sqrt{\rho}\, e_n\|^2 = \operatorname{tr}\rho = 1$. The derivative is 0 in operator norm, for every density operator.
- No phase choice: if unitaries $V(t)$ induce Φ_t on pure states near 0, then $V(t) = c(t)U(t)$ with $c(t) = \langle e_0|V(t)e_0\rangle$ and $|c(t)| = 1$, since $U(t)e_0 = e_0$. If V is norm-differentiable at 0, so are c and $U = c^{-1}V$. For each n, $U(t)e_n = e_n$ near $t = 0$, so the derivative $U'(0)$ vanishes on every e_n, hence $U'(0) = 0$. But $\|U(t) - 1\| \ge \|(U(t) - 1)e_{k(t)}\| = |e^{it} - 1|$, and $|e^{it} - 1|/|t| \to 1$, so $U(t) - 1$ is not $o(t)$.

Notes.

- The equation still holds on vectors: $U(t)\psi$ has derivative 0 at 0 for every ψ, so $H = 0$ in `schrodinger_of_preserves_vonNeumannEntropy`. What fails is only the operator-norm form $U'(t_0) = (i\hbar)^{-1} H U(t_0)$, which the finite-dimensional `schrodinger_of_preserves` gives.
- EVDIFF holds here only at $t_0 = 0$: $k(t)$ jumps at $t = \pm 1/n$, and so does $U(t)$. A smooth variant (not formalized) puts the phase $\theta_k(t) = \varphi(kt)/k$, for a smooth bump φ supported

in $[1,2]$ with $\varphi(3/2) \neq 0$, on every e_k at once. Near each $t \neq 0$ only finitely many θ_k are nonzero, so $U(t)$ is norm-differentiable there; at 0, $\|\Phi_t(\rho) - \rho\| \leq 2\,\|U(t) - 1\|\,\|\rho Q_t\|$ with Q_t the projection onto $\overline{\operatorname{span}}\{e_k : k \geq 1/|t|\}$, and $\|\rho Q_t\| \to 0$ since ρ is compact. So EVDIFF holds at every time, also for the two-parameter family $U(t_1)U(t_0)^\dagger$, while $\|U(t)-1\| \geq |e^{i\varphi(3/2)/k}-1|$ at $t = 3/(2k)$, which is again not $o(t)$.

- With EVTI the propagator is $e^{t(i\hbar)^{-1}H}$ with bounded H, which is norm-differentiable; the example needs a time-dependent evolution.

Problems found in the brief.

- Step 3 differentiates $U(t)$ as an operator. In infinite dimension EVDIFF, even in operator norm on every density operator, does not give a norm-differentiable $U(t)$, only a strongly differentiable one; Step 3 then goes through on vectors, as in Section 1.2.

1.6 Unbounded Hamiltonians

Files.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/Unbounded.lean`
- Imports: `SchroedingerEvolution.InfiniteDimensional`, the any-dimension part of the brief "Multiple equivalent characterizations of the Schrödinger equation" (for `IsEvolution`, `DrInfo`, `DrSceq` and `schrodinger_tfae`). That module imports `UnitaryEntropy.InfiniteDimensional`, so the module graph stays acyclic, but the library `UnitaryEntropy` now depends on the library `SchroedingerEvolution`.
- File: `briefs/UnitaryEntropy/UnitaryEntropy/MomentumExtensions.lean`, the example in "Problems found in the brief" below. Imports: `ForMathlib.MeasureTheory.Function.LpSpace.Multiplication` (multiplication operators and dominated convergence in L^p, Section 1.4), `Mathlib.Analysis.Fourier.AddCircle` (the Fourier basis of L^2 of the circle), `Mathlib.Analysis.InnerProductSpace.LinearPMap` (adjoints of unbounded operators), `Mathlib.Analysis.InnerProductSpace.Calculus`, `Mathlib.Analysis.SpecialFunctions.Integrals.Basic` and `Mathlib.MeasureTheory.Function.Floor`.
- File: `briefs/UnitaryEntropy/UnitaryEntropy/MomentumDomain.lean`, the identification of D with $H^1_0(0,1)$ in that example. Imports: `UnitaryEntropy.MomentumExtensions`, `ForMathlib.Analysis.InnerProductSpace.HilbertBasisMaps` (coordinates in a Hilbert basis and Parseval), `Mathlib.Analysis.Normed.Group.Tannery` (dominated convergence for sums) and the new `ForMathlib.MeasureTheory.Integral.IntervalIntegral.IntegrationByParts`.
- New `ForMathlib` file `ForMathlib/MeasureTheory/Integral/IntervalIntegral/IntegrationByParts.lean`: Fubini on the triangle $a \leq v \leq u \leq b$, $\int_a^b k(u) \int_a^u G\,du = \int_a^b G(v) \int_v^b k\,dv$ (`intervalIntegral.integral_mul_primitive`), and from it integration by parts against a primitive, $\int_a^b K'\Phi = K(b)\Phi(b) - \int_a^b KG$ for $\Phi(u) = \int_a^u G$ with G only integrable (`intervalIntegral.integral_deriv_mul_primitive`). It imports `Mathlib.MeasureTheory.Integral.IntervalIntegral.FundThmCalculus` and `Mathlib.MeasureTheory.Integral.Prod`.

- `ForMathlib/Analysis/InnerProductSpace/HilbertBasisMaps.lean` gains `HilbertBasis.map`, the image $i \mapsto L(b_i)$ of a Hilbert basis under a linear isometric equivalence L.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/MomentumPropagator.lean`, the momenta H_α and $U_\alpha(t) = e^{-itH_\alpha/\hbar}$ in that example. Imports: `UnitaryEntropy.MomentumDomain` and `ForPhyslib.QuantumMechanics.Stone` (for `SolvesSchrodinger`).

- `ForPhyslib/QuantumMechanics/Stone.lean` gains `schrodingerPropagator`, the operator $e^{-itH/\hbar}$ of a self-adjoint H given by the spectral theorem, as $\int e^{it\lambda}\, dP(\lambda)$ for the spectral measure P of $-H/\hbar$, and `SolvesSchrodinger.eq_schrodingerPropagator`: a family of bounded operators with $U(0) = I$ that solves the Schrödinger equation for H is $e^{-itH/\hbar}$. The converse of Stone's theorem, `exists_stronglyContinuousUnitaryGroup_of_solvesSchrodinger`, now follows from it.

- File: `briefs/UnitaryEntropy/UnitaryEntropy/VaryingPhase.lean`, the example with a continuously varying boundary phase (last item below). Imports: `ForPhyslib.QuantumMechanics.ForwardHamiltonian`, `UnitaryEntropy.MomentumPropagator`.

- New file `ForPhyslib/QuantumMechanics/ForwardHamiltonian.lean`: the forward Hamiltonian of any unitary evolution, and a local Stone theorem ("A local Stone theorem" below). Imports: `ForPhyslib.QuantumMechanics.Stone`, to which `solvesSchrodinger_schrodingerPropagator` ($e^{-itH/\hbar}$ solves the Schrödinger equation for H), `schrodingerPropagator_zero` and `schrodingerPropagator_mem_unitary` were added.

- New file `ForPhyslib/QuantumMechanics/Howland.lean`: Howland's extended space, the positive counterpart ("Howland's generator" below). Imports: `ForPhyslib.QuantumMechanics.Stone`, `ForMathlib.MeasureTheory.Function.LpSpace.Multiplication` (dominated convergence in L^p) and TauCeti's `TauCeti.MeasureTheory.Function.Lp.Translation` (strong continuity of translation in L^p).

- New file `ForPhyslib/QuantumMechanics/HowlandGenerator.lean`: the Howland generator computed on compactly supported functions (end of "Howland's generator" below). Imports: `ForPhyslib.QuantumMechanics.ForwardHamiltonian`, `ForPhyslib.QuantumMechanics.Howland`.

- New file `briefs/UnitaryEntropy/UnitaryEntropy/HowlandPropagator.lean`: Howland's construction for the propagator of DR-INFO + EVDIFF. Imports: `UnitaryEntropy.TwoParameter`, `ForPhyslib.QuantumMechanics.HowlandGenerator`. To `Howland.lean` was added `UnitaryPropagator.ofContinuous`, the propagator $U(t)U(s)^\dagger$ of a norm-continuous family of unitaries.

Setting. As in Section 1.2, restricted to a time-independent evolution (EVTI): Φ_t for $t \geq 0$, with $\Phi_0 = I$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$, affine on the density operators and continuous at 0^+ in operator norm on each density operator (`IsEvolution`). DR-INFO is entropy preservation together with a reconstructed past for every density operator (`DrInfo`). DR-SCEQ is conjugation by a propagator U with $U(0) = I$ that solves $\frac{d}{dt}U(t)\psi = (i\hbar)^{-1}HU(t)\psi$ for ψ in the domain of a self-adjoint, in general unbounded, H (`DrSceq`). EVDIFF is weakened to a dense set of states (`EvDiff`): there is a dense subspace D such that $t \mapsto \Phi_t(|\psi\rangle\langle\psi|)$ is differentiable on $t \geq 0$, in operator norm, for every unit vector $\psi \in D$.

Results.

Brief	Lean
EVDIFF on a dense set of states	`EvDiff`
Theorem 2 ($\Rightarrow$), EVDIFF	`evDiff_of_drSceq` (with `differentiableAt_proj_of_hasDerivAt`)
Theorem 2 with EVTI, unbounded H	`drSceq_iff_drInfo_and_evDiff`
Without EVTI: the twisted translations $U_\alpha(t)$ are unitary	`twistedTranslation`, `twistedTranslation_mem_unitary`
Without EVTI: $H_\alpha = -i\,d/dx$ with $\psi(1) = e^{i\alpha}\psi(0)$ is self-adjoint	`momentum`, `isMomentumPair_iff`, `isSelfAdjoint_momentum`
Without EVTI: $U_\alpha(t) = e^{-itH_\alpha/\hbar}$	`solvesSchrodinger_twistedTranslation`, `twistedTranslation_eq_schrodingerPropagator`
Without EVTI: $U(t)\psi$ is differentiable at 0 on a dense D	`switchedPropagator`, `switchedDomain`, `dense_switchedDomain`
Without EVTI: $H_0 = i\hbar U'(0)$ on D is symmetric but not self-adjoint	`switchedHamiltonian_symm`, `not_isSelfAdjoint_switchedHamiltonian`, `switchedHamiltonian_counterexample`
Without EVTI: $D = H_0^1(0,1)$ for $e^{i\alpha} \neq e^{i\beta}$	`sobolevZero`, `switchedDomain_eq`
Without EVTI: $H_0 = -i\,d/dx$ on D, the minimal momentum	`switchedHamiltonian_primitive`, `exists_primitive_of_mem_switchedDomain`
Without EVTI: the right (left) derivative of $U(t)\psi$ at 0 exists exactly on the domain of H_α (H_β), and $H_0 \subseteq H_\alpha$, $H_0 \subseteq H_\beta$	`exists_hasDerivWithinAt_iff`, `switchedHamiltonian_le_momentum_left`, `switchedHamiltonian_le_momentum_right`
Forward derivative: the propagator of $H(t) = H_{\alpha(t)}$ is unitary, with $U(t_2,t_1)U(t_1,t_0) = U(t_2,t_0)$; for constant α it is $e^{-i(t_1-t_0)H_\alpha}$	`phasePropagator`, `phasePropagator_mem_unitary`, `phasePropagator_comp`, `phasePropagator_const`
Forward derivative: $\lvert\alpha(t_0+s) - \alpha(t_0)\rvert = O(\sqrt{s})$ gives $H_0 \subseteq H_{\alpha(t_0)}$, and $o(\sqrt{s})$ gives $H_0 = H_{\alpha(t_0)}$	`forwardHamiltonian`, `forwardHamiltonian_le_momentum`, `forwardHamiltonian_eq_momentum`
Forward derivative: $H_0^1(0,1)$ is in the forward domain, for every α	`hasDerivWithinAt_phasePropagator_primitive`
Forward derivative, $\alpha(t) = \alpha_0 + \sqrt{t}$: at 0 the forward domain is $H_0^1(0,1)$ and H_0 is not self-adjoint; at $t_0 \neq 0$, $H_0 = H_{\alpha(t_0)}$	`forwardDomain_sqrtPhase`, `forwardHamiltonian_sqrtPhase`, `not_isSelfAdjoint_forwardHamiltonian_sqrtPhase`, `sqrtPhase_counterexample`
Forward derivative, any unitary evolution: the forward Hamiltonian is symmetric, and it is self-adjoint iff the evolution is tangent at t_0 to a one-parameter unitary group, which is then $e^{-i\tau H_0/\hbar}$	`QuantumMechanics.forwardHamiltonian`, `forwardHamiltonian_isSymmetric`, `forwardHamiltonian_eq_of_isLittleO`, `isSelfAdjoint_forwardHamiltonian_iff`

Brief	Lean
Forward derivative, $\alpha(t) = \alpha_0 + \sqrt{t}$: at 0 no one-parameter unitary group is tangent to the evolution	`not_exists_tangent_sqrtPhase`
Forward derivative, invariant core: a dense subspace invariant under the evolution, on which the generators near t_0 differ from the one at t_0 by bounded operators, is a core, and the forward Hamiltonian is essentially self-adjoint	`isEssentiallySelfAdjoint_forwardHamiltonian`, `exists_eq_zero_of_inner_deriv`, `norm_inner_le_of_hasDerivAt`
Forward derivative, invariant core with generator differences tending to 0 as $t \to t_0^+$: the forward Hamiltonian is self-adjoint	`isSelfAdjoint_forwardHamiltonian`, `norm_sub_schrodingerPropagator_le`
Howland: a strongly continuous unitary propagator $U(t, s)$ gives a strongly continuous unitary group W on $L^2(\mathbb{R}; E)$ with self-adjoint generator K	`UnitaryPropagator`, `UnitaryPropagator.howlandGroup`, `UnitaryPropagator.exists_howlandGenerator`
Howland: K is the forward Hamiltonian of W; on a compactly supported continuous f whose moving orbit $\sigma \mapsto U(t, t-\sigma)f(t-\sigma)$ has jointly continuous derivative G, $Kf = i\hbar\, G(0, \cdot)$; for $U(t, s) = V(t-s)$, $K(\chi x) = \chi H x - i\hbar\chi' x$	`UnitaryPropagator.howlandGenerator`, `isSelfAdjoint_howlandGenerator`, `howlandGenerator_toLp`, `howlandGenerator_smul_of_group`
Howland, finite dimension: DR-INFO + EVDIFF give a strongly continuous unitary propagator $U(t)U(s)^\dagger$ inducing the evolution, and with it a self-adjoint K	`TwoParameter.exists_unitaryPropagator`, `TwoParameter.exists_isSelfAdjoint_howlandGenerator`

Differences from the brief.

- Only the time-independent case is formalized. For ($\Leftarrow$), DR-INFO gives DR-SCEQ by `schrodinger_tfae`, through DR-PROB, DR-EV, DR-UNIT and Stone's theorem; EVDIFF is not used, since continuity and the group law already give a self-adjoint generator. For ($\Rightarrow$), DR-INFO is part of `schrodinger_tfae`, and the dense subspace in EVDIFF is the domain of H: for ψ there, $t \mapsto U(t)\psi$ is differentiable, hence so is $t \mapsto |U(t)\psi\rangle\langle U(t)\psi| = \Phi_t(|\psi\rangle\langle\psi|)$, as $(x, y) \mapsto |x\rangle\langle y|$ is a bounded real-bilinear map.
- The time-dependent case of Theorem 2 with unbounded $H(t)$ is not formalized. The examples in the next paragraph, showing that its ($\Leftarrow$) direction fails without further hypotheses, with the two-sided and with the forward derivative, are.

Problems found in the brief.

- For unbounded generators, Step 3 gives only a symmetric $H(t_0)$. If $t \mapsto U(t)\psi$ is differentiable at t_0 only for ψ in a dense subspace D, differentiating $\langle U(t)\psi|U(t)\phi\rangle = \langle\psi|\phi\rangle$ gives $\langle A\psi|\phi\rangle =$

$-\langle\psi|A\phi\rangle$ on D: $A(t_0)$ is skew-symmetric, which is weaker than $A(t_0) = -A(t_0)^\dagger$ for an unbounded $A(t_0)$. Then $H(t_0)$ need not be self-adjoint, even on the largest such D. Example (`switchedHamiltonian_counterexample`): on $L^2(0,1)$ let H_α and H_β be $-i\,d/dx$ with the boundary conditions $\psi(1) = e^{i\alpha}\psi(0)$ and $\psi(1) = e^{i\beta}\psi(0)$, $\alpha \neq \beta$, both self-adjoint. Take $H(t) = H_\beta$ for $t < 0$ and $H(t) = H_\alpha$ for $t \geq 0$, so the family differentiated at $t_0 = 0$ is $U(t) = U_{0\to t} = e^{-itH_\alpha/\hbar}$ for $t \geq 0$ and $U(t) = U_{t\to 0}^\dagger = e^{-itH_\beta/\hbar}$ for $t \leq 0$. The evolution is unitary, so it satisfies [B]EVMIX and DR-INFO. The right derivative of $U(t)\psi$ at 0 exists exactly for ψ in the domain of H_α, the left one for ψ in the domain of H_β, so $U(t)\psi$ is differentiable at 0 exactly when $\psi \in H^1(0,1)$ with $\psi(0) = \psi(1) = 0$. This subspace is dense, but $-i\,d/dx$ on it is the minimal momentum operator, closed and symmetric with deficiency indices $(1,1)$, and not self-adjoint. So the self-adjointness of $H(t)$ has to be assumed, as in Kato's theory of time-dependent Hamiltonians, rather than derived from EVDIFF. With EVTI it is derived, by Stone's theorem.

The formalization works on L^2 of the circle $\mathbb{R}/\mathbb{Z}$ with Haar measure, which is $L^2(0,1)$ through the coordinate $x \in [0,1)$ (`circleRep`). $U_\alpha(t)$ is defined directly as the twisted translation $\psi(x) \mapsto e^{i\alpha k}\psi(x - t/\hbar)$, with k the number of wrap-arounds (`twistedTranslation`); that it is $e^{-itH_\alpha/\hbar}$ is proved separately (last item below). D is the set of ψ for which $t \mapsto U(t)\psi$ is differentiable at 0 (`switchedDomain`), and $H_0\psi = i\hbar\,U'(0)\psi$ on D (`switchedHamiltonian`). What is proved:

- D is dense. It contains $e^{2\pi inx} - 1$ for every n (`hasDerivAt_twistedTranslation_fourierDiff`): this function vanishes at both ends, so its quasi-periodic extension $e^{i\gamma\lfloor y\rfloor}(e^{2\pi iny} - 1)$ is Lipschitz for every γ, and the difference quotients converge in L^2 by dominated convergence, to the same limit for α and β. A vector orthogonal to all of them has all Fourier coefficients equal, and the coefficients are square-summable, so it is 0.
- H_0 is symmetric, from $\langle U(t)\psi|U(t)\phi\rangle = \langle\psi|\phi\rangle$.
- H_0 is not self-adjoint. For the plane wave $e^{i\alpha x}$, $U_\alpha(t)e^{i\alpha x} = e^{-i\alpha t/\hbar}e^{i\alpha x}$ and $\langle e^{i\alpha x}|U_\alpha(t)\psi\rangle = e^{-i\alpha t/\hbar}\langle e^{i\alpha x}|\psi\rangle$. So $\langle e^{i\alpha x}|H_0\psi\rangle = \alpha\langle e^{i\alpha x}|\psi\rangle$ on D, and $e^{i\alpha x}$ lies in the domain of $H_0^\dagger$ (`planeWave_mem_adjoint_domain`). But $t \mapsto \langle e^{i\beta x}|U(t)e^{i\alpha x}\rangle$ has the right derivative $-i\alpha c/\hbar$ and the left derivative $-i\beta c/\hbar$ at 0, with $c = \int_0^1 e^{i(\alpha-\beta)x}\,dx \neq 0$, so $e^{i\alpha x} \notin D$ (`planeWave_not_mem_switchedDomain`). Hence $H_0^\dagger \neq H_0$.
- $D = H_0^1(0,1)$ when $e^{i\alpha} \neq e^{i\beta}$ (`switchedDomain_eq`), and $H_0\psi = -i\psi'$ there (`switchedHamiltonian_primitive`, `exists_primitive_of_mem_switchedDomain`). Mathlib has no Sobolev spaces, so $H_0^1(0,1)$ is taken in its absolutely continuous form (`sobolevZero`): the $\psi(x) = \int_0^x g$ with $g \in L^2(0,1)$ and $\int_0^1 g = 0$, so that $\psi(0) = \psi(1) = 0$. The proof uses, for each γ, the Hilbert basis $e_n = e^{i(\gamma+2\pi n)x}$, $n \in \mathbb{Z}$ (`twistedBasis`). It consists of eigenvectors of $U_\gamma(t)$ with eigenvalues $e^{-i(\gamma+2\pi n)t/\hbar}$. So $t \mapsto U_\gamma(t)\psi$ has derivative $-(i/\hbar)h$ at 0 iff $\langle e_n|h\rangle = (\gamma + 2\pi n)\langle e_n|\psi\rangle$ for all n. The coefficient relation is necessary already for a one-sided derivative (`inner_twistedBasis_of_hasDerivWithinAt`). It is sufficient by dominated convergence of the coordinates (`hasDerivAt_twistedTranslation_of_inner`). Integration by parts against the primitive $\Phi(x) = \int_0^x g$ gives $\langle e^{i\delta x}| - ig\rangle = \delta\langle e^{i\delta x}|\Phi\rangle - ie^{-i\delta}\Phi(1)$ (`inner_planeWave_neg_I_smul`). If $\Phi(1) = 0$, this is the coefficient relation for every γ, so $H_0^1 \subseteq D$. Conversely, let $\psi \in D$ and $g = iH_0\psi$. Then $f = \psi - \Phi$ satisfies $(\gamma + 2\pi n)\langle e_n|f\rangle = -ie^{-i\gamma}\Phi(1)$ for $\gamma = \alpha$ and $\gamma = \beta$. If $e^{i\gamma} = 1$, the term $n = -\gamma/2\pi$ gives $\Phi(1) = 0$. Otherwise f is the constant $\Phi(1)/(e^{i\gamma} - 1)$. Since $e^{i\alpha} \neq e^{i\beta}$, both cases

give $\Phi(1) = 0$ and $f = 0$. For $e^{i\alpha} = e^{i\beta}$ the switch does nothing, and D is the larger domain of H_α.

- $U_\alpha(t) = e^{-itH_\alpha/\hbar}$ (`twistedTranslation_eq_schrodingerPropagator`). H_α is defined by its graph (`momentum`): the pairs $(\psi, -ig)$ with $\psi(x) = c + \int_0^x g$, $g \in L^2(0,1)$ and $c + \int_0^1 g = e^{i\alpha}c$, which is $\psi(1) = e^{i\alpha}\psi(0)$. By the integration by parts above, (ψ, h) is in the graph iff $\langle e_n|h\rangle = (\alpha + 2\pi n)\langle e_n|\psi\rangle$ for all n, with $e_n = e^{i(\alpha+2\pi n)x}$ (`isMomentumPair_iff`). So H_α is diagonal in this basis with real eigenvalues, hence self-adjoint (`isSelfAdjoint_momentum`). $U_\alpha(t)$ is diagonal in the same basis, with eigenvalues $e^{-i(\alpha+2\pi n)t/\hbar}$. So it maps the domain of H_α to itself, and the sufficiency of the coefficient relation gives $\frac{d}{dt}U_\alpha(t)\psi = (i\hbar)^{-1}H_\alpha U_\alpha(t)\psi$ there (`solvesSchrodinger_twistedTranslation`). A solution of the Schrödinger equation with $U(0) = I$ is the spectral-theorem exponential $e^{-itH/\hbar}$ (`SolvesSchrodinger.eq_schrodingerPropagator`). The necessity of the coefficient relation, already for one-sided derivatives, shows that the right derivative of $U(t)\psi$ at 0 exists exactly for ψ in the domain of H_α, and the left one exactly for ψ in the domain of H_β (`exists_hasDerivWithinAt_iff`); on D, $H_0 = H_\alpha = H_\beta$ (`switchedHamiltonian_le_momentum_left`, `switchedHamiltonian_le_momentum_right`).

- The forward derivative does not help. The example above uses the two-sided derivative at t_0, and each one-sided derivative alone has a self-adjoint generator. The evolution only runs forward, so $H(t_0)$ can be defined from the right derivative of $t \mapsto U(t, t_0)\psi$ alone (`forwardHamiltonian`, with $\hbar = 1$). It still need not be self-adjoint, for a continuous family of self-adjoint $H(t)$ (`sqrtPhase_counterexample`). Take $H(t) = H_{\alpha(t)}$ with $\alpha(t) = \alpha_0 + \sqrt{t}$ (α_0 for $t \le 0$). Its propagator translates by $t_1 - t_0$, and a point that wraps around at the time s picks up the phase $e^{-i\alpha(s)}$ (`phasePropagator`): it is unitary, satisfies $U(t_2,t_1)U(t_1,t_0) = U(t_2,t_0)$, and is $e^{-i(t_1-t_0)H_\gamma}$ for constant $\alpha = \gamma$ (`phasePropagator_const`). What is proved:

  - For $\tau < 1$, $U(t_0 + \tau, t_0)$ differs from $e^{-i\tau H_\gamma}$, $\gamma = \alpha(t_0)$, only on the strip $0 \le x < \tau$ of points that have wrapped around. There the phase is $e^{-i\alpha(t_0+\tau-x)}$ instead of $e^{-i\gamma}$, applied to $\psi(x - \tau + 1)$ (`phasePropagator_sub_twistedTranslation_eq`). Against the basis vector e_n, the difference is at most $\sup|\alpha - \gamma|\,\sqrt{\tau}\,\|\psi\|_{L^2(1-\tau,1)}$ (`norm_inner_planeWave_mulL_stripPhase_le`).
  - If $|\alpha(t_0 + s) - \alpha(t_0)| = O(\sqrt{s})$ for $s \ge 0$, this is $o(\tau)$. So a right derivative v gives the coefficient relation of H_γ, and $H_\gamma\psi = iv$: the forward Hamiltonian is a restriction of H_γ (`forwardHamiltonian_le_momentum`). If it is $o(\sqrt{s})$, every ψ in the domain of H_γ is bounded, the whole difference is $o(\tau)$, and the forward Hamiltonian is H_γ (`forwardHamiltonian_eq_momentum`). For $\alpha_0 + \sqrt{t}$ this holds at every $t_0 \ne 0$ (`sqrtPhase_little0`): there the forward Hamiltonian is the self-adjoint $H_{\alpha(t_0)}$.
  - For every α, $H_0^1(0,1)$ is in the forward domain, with $H_0\psi = -i\psi'$ (`hasDerivWithinAt_phasePropagator_primitive`). On H_0^1 the tail $\|\psi\|_{L^2(1-\tau,1)}$ is $o(\tau)$: it is half the norm of $e^{-i\tau H_{\gamma+\pi}}\psi - e^{-i\tau H_\gamma}\psi$, and both terms have the derivative $-\psi'$.
  - At $t_0 = 0$ the forward domain is $H_0^1(0,1)$ (`forwardDomain_sqrtPhase`). A ψ in it is in the domain of H_{α_0}, so $\psi(x) = c + \int_0^x g$ is continuous with $\psi(1) = e^{i\alpha_0}\psi(0)$. On the strip $x < \tau/2$, $|e^{-i\sqrt{\tau - x}} - 1| \ge (2/\pi)\sqrt{\tau/2}$ and $|\psi(x - \tau + 1)| \ge |\psi(1)|/2$ for small τ, so $\|U(\tau,0)\psi - e^{-i\tau H_{\alpha_0}}\psi\| \ge |\psi(1)|\tau/2\pi$. Both terms have the right derivative v, so the difference is $o(\tau)$, and $\psi(1) = \psi(0) = 0$ (`mem_sobolevZero_of_hasDerivWithinAt_sqrtPhase`). The forward Hamiltonian at 0 is then the minimal momentum $-i\,d/dx$ on

$H_0^1(0,1)$, the operator H_0 of the switched example (`forwardHamiltonian_sqrtPhase`): dense domain, symmetric, not self-adjoint (`not_isSelfAdjoint_forwardHamiltonian_sqrtPhase`).

So $\frac{1}{2}$ is the threshold: Hölder continuity of α with exponent above $\frac{1}{2}$ gives the self-adjoint $H_{\alpha(t_0)}$, exponent $\frac{1}{2}$ does not.

Howland's generator. What survives without any hypothesis on $H(t)$ is a self-adjoint operator on the extended space $L^2(\mathbb{R}; E)$, with time as a coordinate (Howland 1974; Reed–Simon II, Section X.12). Let $U(t,s)$ be a strongly continuous unitary propagator: $U(t,t) = I$, $U(t,r)U(r,s) = U(t,s)$, and $(t,s) \mapsto U(t,s)x$ continuous for every x (`UnitaryPropagator`). Then $(W(\sigma)f)(t) = U(t,t-\sigma)f(t-\sigma)$ is a strongly continuous unitary group (`UnitaryPropagator.howlandGroup`), so by Stone's theorem $W(\sigma) = e^{-i\sigma K/\hbar}$ for a self-adjoint K (`UnitaryPropagator.exists_howlandGenerator`), formally $K = -i\hbar\,\partial_t + H(t)$.

- $W(\sigma)$ is translation by σ (`shiftOp`) followed by the pointwise operator $f(t) \mapsto U(t,t-\sigma)f(t)$ (`UnitaryPropagator.diagOp`). The latter is strongly measurable because $(t,x) \mapsto U(t,t-\sigma)x$ is jointly continuous, being strongly continuous and uniformly bounded (`continuous_uncurry_of_norm_map`). Both are isometries.

- The propagator law gives $W(\sigma)W(\rho) = W(\sigma+\rho)$ (`UnitaryPropagator.howland_add`), and $W(0) = I$, so each $W(\sigma)$ is unitary.

- Strong continuity: $W(\sigma)f - f$ is the translation increment $\mathrm{diagOp}(\sigma)(f(\cdot-\sigma)-f)$, which tends to 0 by TauCeti's `MeasureTheory.Lp.continuous_compMeasurePreserving_add_right`, plus $U(t,t-\sigma)f(t) - f(t)$, which tends to 0 by dominated convergence (`UnitaryPropagator.tendsto_howland_zero`).

- A time-independent evolution is the special case $U(t,s) = V(t-s)$ for a strongly continuous unitary group V (`StronglyContinuousUnitaryGroup.toUnitaryPropagator`).

The theorem uses only unitarity and strong continuity of the propagator. In finite dimension, DR-INFO + EVDIFF provide such a propagator, $U(t,s) = U(t)U(s)^\dagger$ (`TwoParameter.exists_propagator`). It is norm-continuous, being differentiable, so `UnitaryPropagator.ofContinuous` makes it a `UnitaryPropagator` (`TwoParameter.exists_unitaryPropagator`), and its Howland generator is self-adjoint with $W(\sigma) = e^{-i\sigma K/\hbar}$ (`TwoParameter.exists_isSelfAdjoint_howlandGenerator`).

K is made concrete as the forward Hamiltonian of W (`UnitaryPropagator.howlandGenerator`): W is tangent to its own group, so by the local Stone theorem K is self-adjoint and $W(\sigma) = e^{-i\sigma K/\hbar}$ (`isSelfAdjoint_howlandGenerator`, `howland_eq_schrodingerPropagator`). It is computed on functions (`howlandGenerator_toLp`): let f be continuous with compact support, and let the moving orbit $\sigma \mapsto U(t,t-\sigma)f(t-\sigma)$ have derivative $G(\sigma,t)$, jointly continuous. Then f is in the domain of K and $Kf = i\hbar\,G(0,\cdot)$.

- $G(0,t) = 0$ off the support of f, so $G(0,\cdot)$ has compact support.

- For $\sigma \in [0,1]$ the difference quotient $(W(\sigma)f - f)/\sigma - G(0,\cdot)$ vanishes outside the compact set $\mathrm{tsupport}\, f + [0,1]$. On it, by the mean value theorem, it is at most $\sup_{s\le\sigma} \|G(s,t) - G(0,t)\|$, uniformly small by joint continuity and compactness. So it tends to 0 in L^2, and the forward derivative exists.

When U and f are differentiable, $G(0,t) = (i\hbar)^{-1}H(t)f(t) - f'(t)$ with $H(t)x = -i\hbar\,\partial_s U(t,s)x|_{s=t}$, so $K = H(t) - i\hbar\,\partial_t$. Formalized in the time-independent case (`howlandGenerator_smul_of_group`): for $U(t,s) = V(t-s)$ with V solving the Schrödinger equation for H, χ compactly supported C^1 and x in the domain of H, $K(\chi x) = \chi\, Hx - i\hbar\chi' x$. The orbit derivative at every σ comes from the group law, $V(s) = V(\sigma)V(s-\sigma)$. Not formalized: a core of K of such functions for a time-dependent family $H(t)$, which needs hypotheses of Kato type, and a relation to Nickel's theorem (an invariant core of K iff U solves the Cauchy problem).

A local Stone theorem. At a single time the forward Hamiltonian is characterized by the evolution alone. Let $F(\tau)$, $\tau \geq 0$, be unitaries with $F(0) = I$, for instance $F(\tau) = U(t_0 + \tau, t_0)$. Its forward Hamiltonian is $H_0\psi = i\hbar\,\partial_+ F(\tau)\psi$ at $\tau = 0$, on the vectors where the right derivative exists (`QuantumMechanics.forwardHamiltonian`; the one of `VaryingPhase.lean` is the case $\hbar = 1$, `forwardHamiltonian_eq_quantumMechanics`).

- H_0 is symmetric (`forwardHamiltonian_isSymmetric`): the right derivative of $\langle F(\tau)\psi, F(\tau)\varphi\rangle = \langle\psi,\varphi\rangle$ at 0 is $\langle\psi',\varphi\rangle + \langle\psi,\varphi'\rangle = 0$.
- H_0 is self-adjoint iff F is tangent at 0 to a one-parameter unitary group: there is a self-adjoint H with $F(\tau)\psi - e^{-i\tau H/\hbar}\psi = o(\tau)$ as $\tau \to 0^+$ for every ψ in the domain of H (`isSelfAdjoint_forwardHamiltonian_iff`). Then $H = H_0$ (`forwardHamiltonian_eq_of_isLittleO`): $e^{-i\tau H/\hbar}\psi$ has the right derivative $(i\hbar)^{-1}H\psi$ on the domain of H (`solvesSchrodinger_schrodingerPropagator`), so $H \subseteq H_0$, and a symmetric extension of a self-adjoint operator is the operator itself, $H \subseteq H_0 \subseteq H_0^\dagger \subseteq H^\dagger = H$ (`IsSelfAdjoint.eq_of_le_of_isSymmetric`). Conversely, take $H = H_0$.
- For $\alpha(t) = \alpha_0 + \sqrt{t}$, no one-parameter unitary group is tangent to the evolution at 0 (`not_exists_tangent_sqrtPhase`), since H_0 is not self-adjoint there.

The criterion is a hypothesis on the evolution, not on a family $H(t)$, but it asks for the tangent group to be given. A condition on F alone that gives essential self-adjointness is also formalized (`isEssentiallySelfAdjoint_forwardHamiltonian`). Let D be a dense subspace with $F(t)D \subseteq D$ for all $t \in \mathbb{R}$, on which $t \mapsto F(t)\psi$ is differentiable, and let $B\psi = \partial_t F(t)\psi|_{t=0}$. Suppose that for $|t| \leq T$ and $\psi \in D$,

$$\|\partial_t F(t)\psi - BF(t)\psi\| \leq C\|\psi\|,$$

that is, the generators near t_0 differ from the one at t_0 by bounded operators on D. Then the restriction of H_0 to D is essentially self-adjoint, and H_0 is essentially self-adjoint with the same closure.

- Let y satisfy $\langle y, Bx\rangle = a\langle y, x\rangle$ for $x \in D$, with $a \geq 0$. Along the orbit $\eta(\tau) = F(\tau)\psi$, which stays in D, $f(\tau) = \langle y, \eta(\tau)\rangle$ has $f' = af + \langle y, e\rangle$ with $\|e\| \leq C\|\psi\|$, so $e^{-a\tau}f(\tau)$ changes by at most $C\|\psi\|\|y\|$ per unit time, and $|f| \leq \|y\|\|\psi\|$. Hence $|\langle y,\psi\rangle| \leq (e^{-aS} + CS)\|y\|\|\psi\|$ for $0 < S \leq T$ (`norm_inner_le_of_hasDerivAt`). For $a < 0$ the same holds backward in time.
- Taking S with $CS \leq \frac{1}{4}$ and $|a| \geq 4/S$ gives $|\langle y,\psi\rangle| \leq \frac{1}{2}\|y\|\|\psi\|$ on D, hence everywhere by density, and $y = 0$ (`exists_eq_zero_of_inner_deriv`).
- For $\kappa = 1/(\mu|\hbar|)$, the deficiency vectors of $\kappa H_0|_D$ at $\pm i$ are such y with $a = \pm\mu$, so both defect numbers vanish and $\kappa H_0|_D$ is essentially self-adjoint; so is $H_0|_D$. Its closure is self-adjoint, and the closure of the symmetric extension H_0 is a symmetric extension of it, hence equal.

No self-adjointness at any time is assumed, and the evolution need not satisfy a composition law. In the example $\alpha_0 + \sqrt{t}$ no dense subspace of the forward domain at 0 is invariant: $U(t,0)\psi$ takes the value $\psi(1-t)$ at 0 up to a phase, and the forward domain is $H^1_0(0,1)$.

If moreover the bound tends to 0 forward in time, that is, for every $\varepsilon > 0$ there is $\delta > 0$ with $\|\partial_t F(t)\psi - BF(t)\psi\| \le \varepsilon\|\psi\|$ for $t \in [0,\delta]$ and $\psi \in D$, then H_0 itself is self-adjoint, and equal to the closure of its restriction to D (`isSelfAdjoint_forwardHamiltonian`). This is the case $H(t) = H + V(t)$ with V bounded and norm-continuous.

- Duhamel estimate (`norm_sub_schrodingerPropagator_le`): if A is self-adjoint and $\eta' = (i\hbar)^{-1}A\eta + e$ with $\|e\| \le R$ on $[0,\tau]$, then $\|\eta(\tau) - e^{-i\tau A/\hbar}\eta(0)\| \le R\tau$. For z in the domain of A, $\langle e^{-isA/\hbar}z, \eta(s)\rangle$ has derivative $\langle e^{-isA/\hbar}z, e(s)\rangle$, since A is symmetric; the bound follows by density of the domain and unitarity.

- With $A = \bar{H}_0$ and $\eta(s) = F(s)\psi$, $\psi \in D$, this gives $\|F(\tau)\psi - e^{-i\tau A/\hbar}\psi\| \le \varepsilon\tau\|\psi\|$ for $\tau \le \delta$; by density for every $\psi \in E$. So F is tangent to $e^{-i\tau A/\hbar}$, and by the local Stone theorem $H_0 = A$.

Not formalized: a relative bound $C(\|B\psi\|+\|\psi\|)$ in place of $C\|\psi\|$, which time-dependent unbounded potentials need.